EC 551 Advanced Digital Design with Verilog and FPGAs

Fall 2026, Prof. Douglas Densmore

Homework #3

Due: 11/3

Synthesis and Scheduling – FAIR GAME FOR THE 2nd EXAM!

1. Consider the following equations:

   w = a – d
   x = a + b
   a = w * x
   x = w + d
   w = d – x
   a = w * b

   a. Create the "single assignment" form of these equations.
   b. Create a data flow graph from the single assignment equations in part a.
   c. Partition (and draw out) the dataflow graph into an ASAP schedule. Partition the dataflow graph into an ALAP schedule (where no cycle is idle but the design is completely sequential with one operation per cycle). Partition the graph into a "register consolidation" schedule (another form of ALAP scheduling yet operations happen in parallel). How many total cycles are required assuming each operation takes 1 cycle?
   d. How many functional units are required for each of these schedules (e.g. how many adders, subtractors, and multipliers)?
   e. If we wanted to synthesize this into hardware, what are some of the next steps? Is it realistic to assume 1 cycle per operation? If not, how would that change the design?

ASM Design – FAIR GAME FOR THE 2nd EXAM!

2. Do problems **5.43, 5.48, 6.5** and **6.7** in the book. *Only create the ASM chart*. You **do not** have to create Verilog code for these. Please ensure the ASM is organized and clearly labeled. Points will be deducted if they are hard to follow or ambiguous.

3. Paper Reading (**available on Blackboard**):

Read **"MIS: A Multiple-Level Logic Optimization System"**

a. What is a "multi-level logic expression"? Why do they exist? Why might they be advantageous over two-level logic expressions?
b. What is the purpose of this paper (**In your OWN words**)?
c. What is a "factored form"? Provide an example **NOT** in the paper.
d. What are "Algebraic and Boolean division"? Provide examples of each that are **NOT** in the paper.
e. What is a "kernel and a co-kernel"? What does the level of a kernel denote? Give examples of kernel and co-kernels. Specify their level as well. These should **NOT** be in the paper.